

FALCON: LandslideGuard AI

Comprehensive System Engineering, AI Models & Risk Fusion Specification

Version: 1.1.0 | Team: Syntax Terrors | Stack: PyTorch, SegFormer, XGBoost, FastAPI, React 18, MQTT, Leaflet

1. Executive Overview & System Purpose

FALCON LandslideGuard AI is an autonomous, real-time landslide monitoring, hazard detection, and early warning intelligence platform. The system solves the vulnerability of high-risk mountain terrains by unifying multi-modal remote sensing and in-situ geotechnical telemetry. By synthesizing high-resolution drone computer vision, 14-band multispectral satellite imagery, live ESP32 IoT environmental sensing over MQTT, meteorological rainfall forecasting, and an advanced 4-tier non-linear geophysical risk engine, FALCON provides zero-latency situational awareness and proactive disaster mitigation.

2. End-to-End System Architecture

The platform utilizes an event-driven, decoupled micro-architecture structured into 4 distinct functional tiers:

Tier / Layer	Technologies & Protocols	Key Responsibilities
1. Edge Sensing & Ingestion	ESP32 Microcontroller, FreeRTOS, MQTT (TCP 1883), HiveMQ Broker	In-situ sampling of soil moisture, pore pressure, air quality (gas), ambient temp, humidity, and ground displacement.
2. Analytical Backend	Python 3.11+, FastAPI, PyTorch (CUDA/CPU), Scikit-Learn, XGBoost, PIL, H5py	Consumes MQTT feeds, executes transformer and tree inference, calculates 4-tier risk fusion, syncs live WeatherAPI & OSM.
3. Real-Time Telemetry Dispatch	FastAPI WebSockets (/ws/live), Asyncio Broadcast Loop, REST API	Pushes live state updates every 2 seconds to all connected dashboard sessions; handles on-demand REST predictions.
4. Presentation & Client UI	React 18, Vite, TypeScript, Tailwind CSS, Zustand, Leaflet, IndexedDB	Glassmorphism interactive HUD, live video feed with bounding overlays, what-if risk simulator, offline snapshot archive.

3. Detailed Specifications of AI & Analytical Models

3.1 Drone Aerial Image Segmentation (FALCON-SegFormer)

- **Architecture:** SegFormer with Mix Transformer encoder backbone (nvidia/mit-b0) + MLP decoder.
- **Weights Location:** backend/models/image/
- **Input:** High-resolution optical RGB aerial drone image frames.
- **Classes:** Binary semantic segmentation (Class 0: Background/Stable Surface, Class 1: Landslide Scarp/Debris Flow).
- **Processing Pipeline:** Images are normalized through AutoImageProcessor, fed through the MiT-B0 encoder, upsampled via bilinear interpolation to original dimensions, and processed through channel-wise softmax.
- **Metrics Generated:** Landslide surface area coverage (%), maximum & mean class probability, total landslide pixel count.
- **Decision Logic:** When landslide coverage exceeds 1.0%, the frame triggers a *LANDSLIDE-LIKE ZONE DETECTED* alert.

3.2 14-Band Satellite Multispectral Segmentation Model

- **Architecture:** 14-channel SegFormer model (nvidia/mit-b0 adapted for 14-channel tensor inputs).
 - **Weights Package:** backend/models/satellite/FALCON_SegFormer_Final.pth (Contains state dict, channel normalization parameters, and metadata).
 - **Input Format:** Geospatial HDF5 (.h5) dataset containing a 3D matrix (Height x Width x 14 bands).
 - **Spectral Bands:** Coastal aerosol, Blue, Green, Red, 3x Vegetation Red Edge bands (705nm, 740nm, 783nm), NIR (842nm), Narrow NIR (865nm), Water Vapour (945nm), SWIR-Cirrus (1375nm), SWIR-1 (1610nm), SWIR-2 (2190nm), and Topographic Digital Elevation Model (DEM) data.
 - **Channel Normalization:** Input arrays undergo zero-mean unit-variance scaling per band using exact training statistics: $(X - \text{Mean}) / (\text{Std} + 1e-8)$.
 - **Hazard Classification:** Area $\geq 20\%$ -> CRITICAL/HIGH, 5-20% -> MODERATE, $>0\%$ -> LOW, 0% -> NONE.

3.3 Geotechnical Numerical Risk Model (XGBoost Hybrid)

- **Architecture:** Trained XGBoost classifier & probability estimator serialized via joblib.
 - **Model Artifact:** backend/models/numerical/FALCON_hybrid_landslide_model.pkl
 - **7-Factor Feature Vector:** [Rainfall_mm, Vegetation_Cover (NDVI), Slope_Angle (deg), Soil_Saturation (%), Temperature_C, Humidity_percent, Soil_Moisture_Content (%)].
 - **Output:** Discrete risk class prediction (0=Stable, 1=Hazard) and calibrated hazard probability $P(\text{High_Risk})$ in [0.0, 1.0].

4. Hierarchical 4-Tier Non-Linear Risk Engine & Fusion Architecture

Rather than simplistic weighted averaging, FALCON uses a 4-tier geophysical decision hierarchy that models slope physics and safeguards against false positives:

Tier	Mechanism	Mathematical & Logical Formulation
Tier 1: Topographic Slope Gate	Non-linear slope factor $G(\theta)$ prevents false alarms on flat terrain.	• $\theta < 15^\circ$: $G(\theta) = \max(0.08, (\theta / 15.0) * 0.20)$ • $15^\circ \leq \theta \leq 45^\circ$: $G(\theta) = \theta / 45.0$ • $\theta > 45^\circ$: $G(\theta) = 1.0$ (Steep escarpment)
Tier 2: Geotechnical ML Inference	Evaluates live telemetry with trained XGBoost model or saturation heuristic.	• Base ML Probability: $P_{ML} = \text{XGBoost}(\text{Telemetry})$ • Fallback Saturation = $0.40 * \text{Soil} + 0.30 * \text{Pore} + 0.30 * \text{Rain}$ • Gated Score = $\text{RawScore} * G(\theta)$
Tier 3: Physical Safety Overrides	Hard fail-safe thresholds immediately elevate the score.	• Shear Displacement ≥ 8.0 mm: Hard floor at 85.0 (CRITICAL) • Shear Displacement ≥ 4.0 mm: Hard floor at 60.0 (HIGH) • Pore Pressure ≥ 50 kPa: Hard floor at 80.0 (CRITICAL) • Rain $\geq 45\text{mm}$ & Soil $\geq 75\%$: Hard floor at 70.0 (HIGH)
Tier 4: Adaptive Vision Fusion	Dynamically balances visual vs geotechnical weights via temporal tracking.	• Confirmed Detection (≥ 2 consecutive frames): $W_{\text{vision}} = 0.45$, $W_{\text{geo}} = 0.55$ • Single Frame Possible (1 frame): $W_{\text{vision}} = 0.30$, $W_{\text{geo}} = 0.70$ • Normal Video Stream: $W_{\text{vision}} = 0.05$, $W_{\text{geo}} = 0.95$ • Camera Offline: $W_{\text{vision}} = 0.00$, $W_{\text{geo}} = 1.00$

5. Hardware, IoT & Protocol Engineering

- **Hardware Nodes:** ESP32 microcontrollers deployed at ground stations or onboard UAV payloads streaming via MQTT.
- **Sensor Array:** DHT22/BME280 (Temperature & Relative Humidity), MQ-135/MQ-4 (Air Quality & Methane/Gas), Geotechnical Geophones & Tensiometers.
- **Broker & Topic:** Connects to broker.hivemq.com:1883 on topic drone/disaster/telemetry.
- **Fault-Tolerant Telemetry Ingestion:** Includes custom pre-parser sanitizing malformed JSON (: nan, : null, broken packets) backed by a regular expression fallback extractor so erratic sensor packets never disrupt the backend.
- **Autonomous Simulation Fallback:** If hardware is offline, a mathematical state generator provides realistic Gaussian-jittered readings so frontend telemetry remains continuous.

6. Meteorological & Geospatial Intelligence

- **Live WeatherAPI Integration:** Queries api.weatherapi.com/v1/forecast.json using the UAV's live GPS coordinates, gathering precipitation rate, temperature, humidity, and 6-hour / 24-hour predictive rainfall accumulation.
- **OSM Nominatim Reverse Geocoding:** Automatically resolves raw latitude/longitude coordinates into human-readable geographic administrative divisions (district, village, mountain ridge).
- **Leaflet Geospatial Mapping:** Visualizes real-time drone position, flight breadcrumbs, alert markers, and terrain contour overlays.

7. API & WebSocket Service Specification

Endpoint	Protocol / Method	Description
/ws/live	WebSocket	Continuous telemetry, risk score & alert broadcast every 2 seconds.
/predict/image	POST (Multipart)	Direct inference on optical drone RGB image with SegFormer model.
/predict/satellite	POST (HDF5 .h5)	Direct inference on 14-band satellite multispectral HDF5 array.
/predict/numerical	POST (JSON)	Direct inference with XGBoost model on 7 geotechnical features.
/predict/risk	POST (Multipart)	Composite multi-modal inference combining image and numerical inputs.
/api/ai/analyze	POST (File/Base64)	Analyzes drone stream frame and immediately updates live system state.
/api/status, /api/sensors, /api/weather, /api/drone	GET (REST)	Standardized REST polling endpoints matching frontend service interfaces.

8. Frontend Architecture & User Interface

- **Framework:** React 18, Vite, TypeScript, Tailwind CSS.
 - **State Management:** Zustand (useTelemetryStore.ts) orchestrating dual-channel communication (primary WebSocket with transparent HTTP polling fallback).
 - **Client-Side Persistence:** IndexedDB powered snapshot repository allowing offline inspection and historical audit trails.
 - **Key Views:**
 1. **Live Monitor:** Live HUD video feed, bounding boxes, optical slip face detection, sensor gauges.
 2. **Risk Engine:** Interactive simulation sliders allowing what-if parameter testing across all 4 tiers.
 3. **AI Analysis:** Dedicated test benches for direct drone image, 14-band satellite HDF5, and geotechnical testing.
 4. **Sensors & Telemetry:** Real-time charts, ESP32 live diagnostics, and threshold triggers.
 5. **Alerts & Snapshots:** Timestamped hazard logs and downloadable snapshot records.